Homework 2
Combinatorics
MATH 31003-01 - Fall 2026

Due: Thursday, September 3, 2026 23:58
Submit: Gradescope, through Blackboard's assignment area.
Please write your homework by hand. Start each exercise on a new page, scan your work to a legible PDF, and clearly mark the pages associated with every exercise.

1. (A constrained integer sum)

Points: 8
How many integer solutions are there to

$$x_1 + x_2 + x_3 + x_4 + x_5 = 19$$

subject to

$$x_1 > 1, \qquad x_2 > 2, \qquad x_3 > 3, \qquad x_4 > 4, \qquad x_5 > 5?$$

2. (A route through specified points)

Points: 12
A northeast lattice path uses only unit steps to the right or unit steps up. How many northeast lattice paths from $(0,0)$ to $(20,10)$ pass through each of the points

$$(5,5), \qquad (10,7), \qquad (15,10)?$$

> **Need a free hint?.** Every required path breaks into four path segments: from $(0,0)$ to $(5,5)$, from $(5,5)$ to $(10,7)$, and so on...

3. (Anagrams with letters and digits)

Points: 10
How many anagrams of `ABRACADABRA676767` have every digit before every letter?

> **Need a free hint?.** First position the digits, then the letters.

4. (Drawing colored balls)

Points: 12
A bag contains an unlimited supply of red balls and blue balls, and exactly 10 yellow balls. A person draws 15 balls.

1. How many possible outcomes are there if the order of the draws is irrelevant? Thus an outcome records only how many balls of each color are drawn.
2. How many possible outcomes are there if the balls are drawn one at a time and placed in a row?

> **Need a free hint?.** For each part, first pretend that yellow balls are unlimited. Then remove the impossible outcomes: ones where you choose more yellow balls than are available.

5. (Passwords with increasing digits)

Points: 10

A password has length 12. Each character is either one of the 26 lowercase letters or one of the decimal digits $0, 1, \ldots, 9$. A password contains exactly three digits; those three digits are distinct and appear in increasing order when read from left to right. The remaining nine characters are letters, which may repeat.

How many passwords satisfy these conditions?

> **Need a free hint?.** Choose the three positions occupied by digits. Next choose the three digit values; their increasing-order requirement then determines which chosen digit goes in each digit position. Finally fill the letter positions.